

Intro and Analog Behavior of Digital Systems

Lecture 01

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Outline

- Course Overview
- Course learning objectives
 - Administrative details
 - Syllabus
 - Schedule
 - Important links and information
 - Labs and Checkoffs
- Final Project
- Embedded Systems Introduction
- Warp Speed Boolean Logic Review

Learning Objectives

Class Norms & Expectations

Instructor

- Be respectful of all students in the class
- Prepare engaging and challenging course material
- Timely feedback to student work and questions

Students

- Be respectful of instructors, grutors, and peers
- Engaged attention in class
- Radical candor

The classroom as an ecosystem

What you do has an impact on those around you, even if you might not realize it.

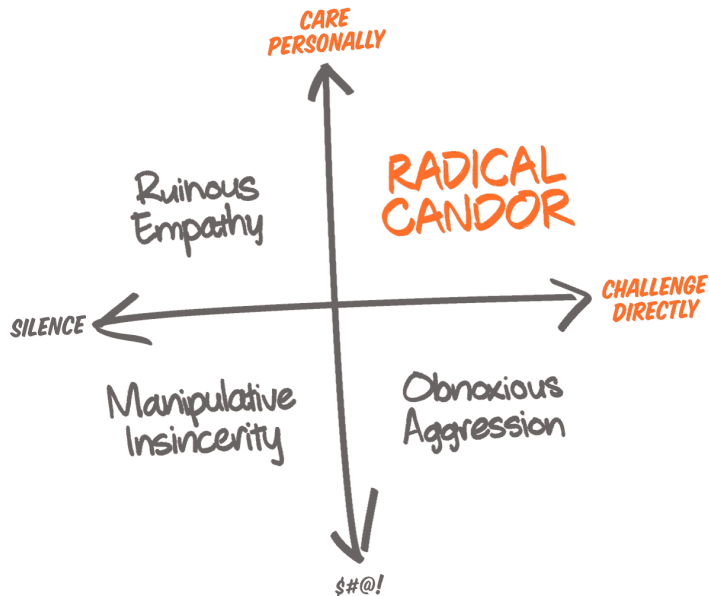
Radical Candor & Psychological Safety

Radical Candor

What is Radical Candor?

Defined by two attributes:

- 1.
- 2.



Psychological Safety

Create an environment where you can courageously challenge yourself.

Failure is ok, it means you're learning!

But there are better and worse ways to fail.

Schedule

Week	Tuesday's Class	Thursday's Class	Due
8/27	Intro & Analog Behavior of Digital Systems	Combinational and Sequential Logic	(Lab Demos)
9/3	Verilog Coding	Synchronous Design	Lab 1 - Development Board Assembly
9/10	FPGA Documentation	Architecture & Assembly Review	Lab 2 - Muxed 7-Segment Display
9/17	Assembly Programming	C Programming on an MCU	Lab 3 - Keypad
9/24	Clock Configuration	Timers	Lab 4 - ARM Assembly Sort
10/1	Serial Interfaces Overview & SPI	UART and the IoT	Lab 5 - Digital Audio
10/8	PCB Design	Advanced Encryption Standard (AES)	Lab 6 - SPI & The Internet of Things
10/15	Happy Fall Break! No class	Project Kickoff	
10/22	Graphics and Displays	Motors and Speakers	Lab 7 - AES & Project Proposal
10/29	Interrupts Pt. 1	Interrupts Pt. 2	Proposal Debriefs
11/5	Presentations	Presentations	
11/12	The Fast Fourier Transform (FFT)	Project Status Report and Demo	Project Status Reports & Demo
11/19	Emerging Topics in Embedded Systems	Happy Thanksgiving! No class	
11/26	Introduction to Real Time	Direct Memory Access	

Important Information

- Course website: <https://pages.hmc.edu/brake/class/e155>
- Discord Server (invite link in welcome email)

Office hours will be scheduled later this week based on clinic meeting schedule.

My Teaching Philosophy

Course Content

Three main sections: labs, project, and in-class participation.

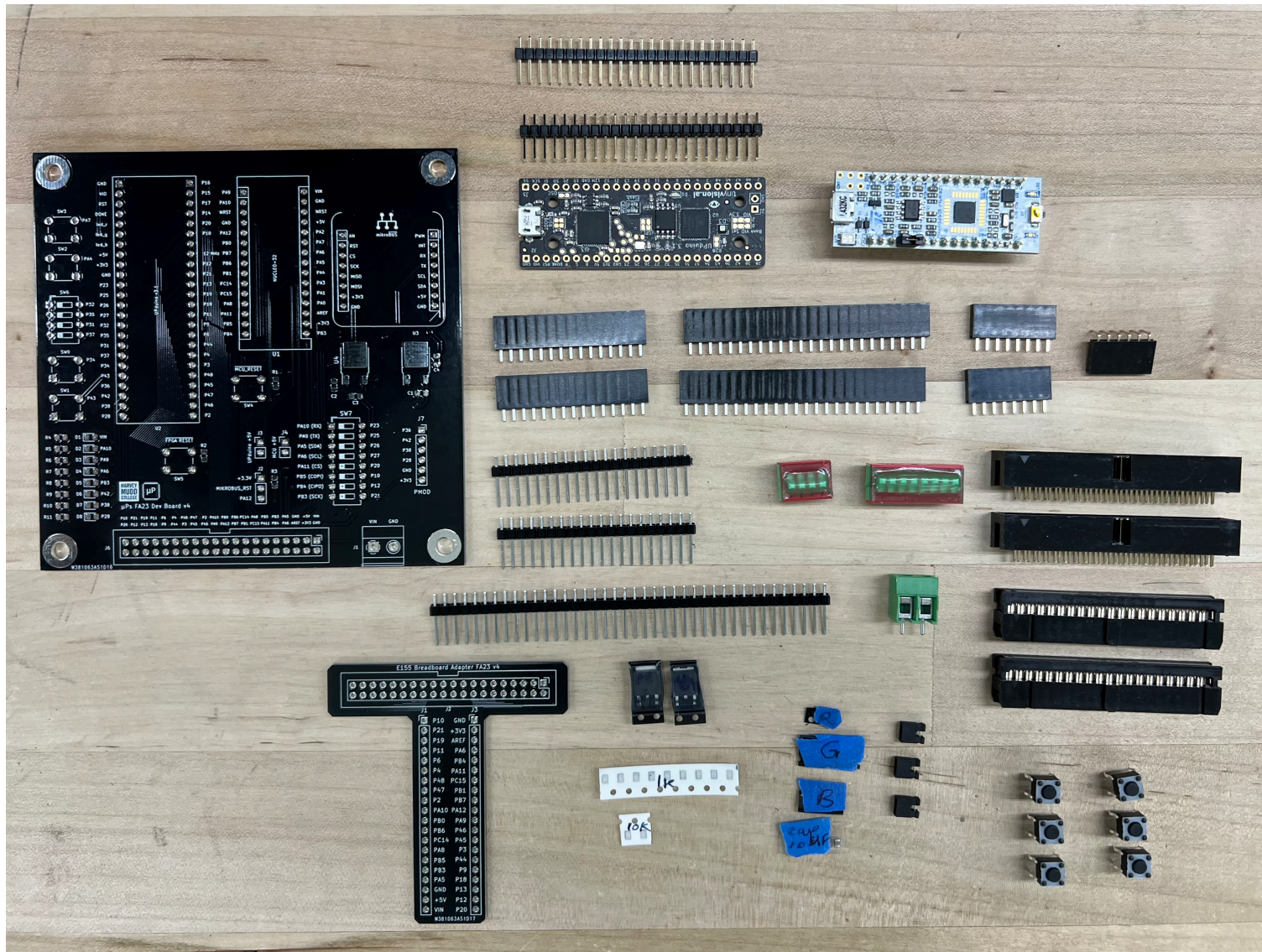
Labs

- Lab 1 - Board Assembly and Testing
- Lab 2 - Multiplexed 7-Segment Display
- Lab 3 - Keypad Scanner
- Lab 4 - ARM Assembly Sort
- Lab 5 - Digital Audio
- Lab 6 - The Internet of Things and Serial Peripheral Interface (SPI)
- Lab 7 - The Advanced Encryption Standard (AES)

Lab Checkoffs

Each week you will demonstrate your lab to an instructor and have it checked off based on that lab's specifications.

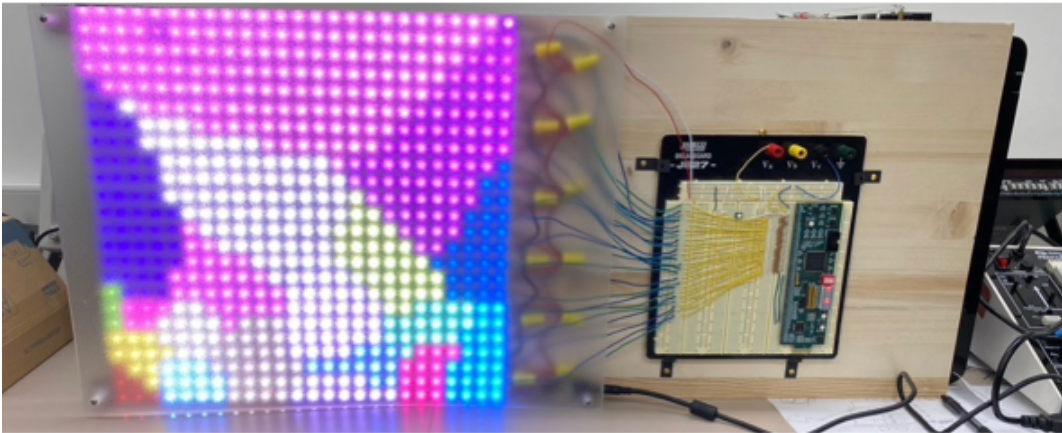
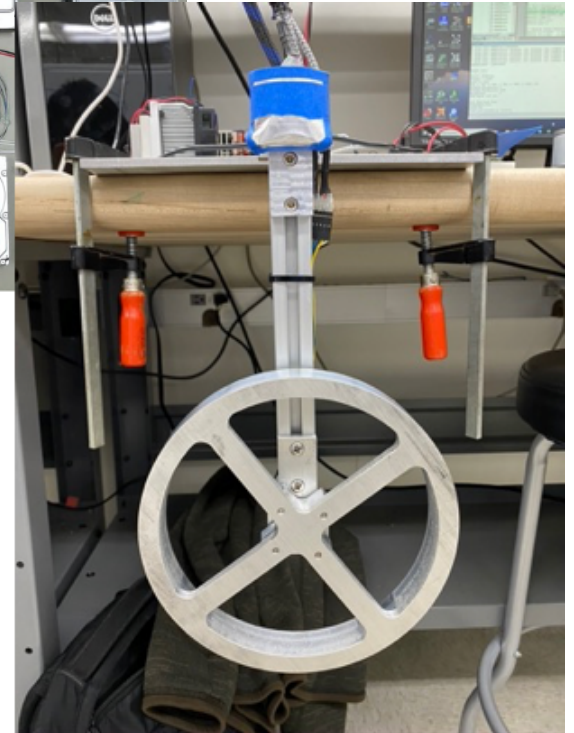
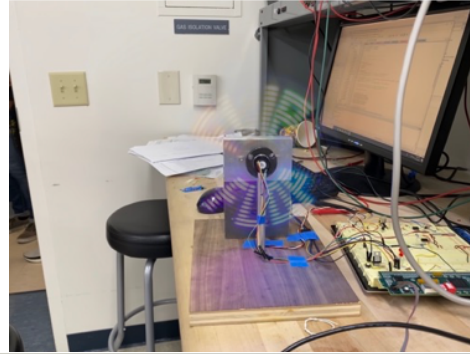
Lab Kit



Project

There are 5 deliverables associated with the project:

1. Proposal
2. Design Review Presentation
3. Mid-point checkoff
4. Final demo
5. Final portfolio website



Grading

What are grades for?

The grading in this course may be a bit different than what you have seen in other courses. Your grade will be determined based on the quality of the deliverables you submit.

Grading

Assignments will be graded on a three point scale designed to reflect your level of mastery of the content contained within it. Every assignment in this class will be presented with a list of specifications with two levels:

1. No Credit
2. Proficiency
3. Mastery

Your grade will be assigned based on whether you meet only the proficiency specs or both the proficiency and mastery specs.

Grading Breakdown

- PS = proficiency specs
- MS = mastery specs

Grade	Labs (7 total)	Project	Class Participation
D	PS < 4	Less than 4 elements meet PS.	More than 3 unexcused absences.
C	PS >= 5. MS > 3/5.	All elements meet PS. 2/5 meet MS.	No more than 3 unexcused absences.
B	PS >= 6. MS > 4/6.	All elements meet PS. 3/5 meet MS.	No more than 2 unexcused absences.
A	PS 7. MS > 5/7.	All elements meet PS. 4/5 meet MS.	No more than 1 unexcused absence.

Practices for Success

- Start early!
- Get stuck, but don't stay stuck.
- Read documentation carefully and repeatedly.
- Ask for help (instructors, grutors, classmates)
- Block your time
- Practice good debugging strategies
 - Follow the signal flow (GIGO)
 - Try a simple case first
 - Use measurement tools to your advantage
 - Use software testing tools like test benches and unit tests.

Embedded Systems Overview

What is an embedded system?

Embedded systems are information processing systems embedded into a larger product

Table 1.2 Distinction between PC-like and embedded system design

	Embedded	PC-/Server-like
Architectures	Frequently heterogeneous very compact	Mostly homogeneous not compact (x86 etc.)
x86 compatibility	Less relevant	Very relevant
Architecture fixed?	Sometimes not	Yes
Model of computation (MoCs)	C+multiple models (data flow, discrete events, ...)	Mostly von Neumann (C, C++, Java)
Optimization objectives	Multiple (energy, size, ...)	Average performance dominates
Safety-critical?	Possibly	Usually not
Real-time relevant	Frequently	Hardly
Applications	Guarantees for several concurrent apps. needed	Best effort approaches to run application
Apps. known at design time	Yes, for real-time systems	Only some (e.g., WORD)

Marwedel, Peter. *Embedded System Design*. Springer, 2018.

What are some common fields where embedded systems are used?

- Transportation
- Factory automation
- Health
- Smart buildings
- Smart grid
- Scientific experiments
- Public safety
- Structural health monitoring
- Disaster recovery
- Robotics
- Agriculture
- Military
- Telecommunication

Challenges

Dependability

- Safety
- Security
- Confidentiality
- Reliability
- Repairability
- Availability

Resource Awareness

- Energy
- Run-time
- Code size
- Weight
- Cost

Design Flow

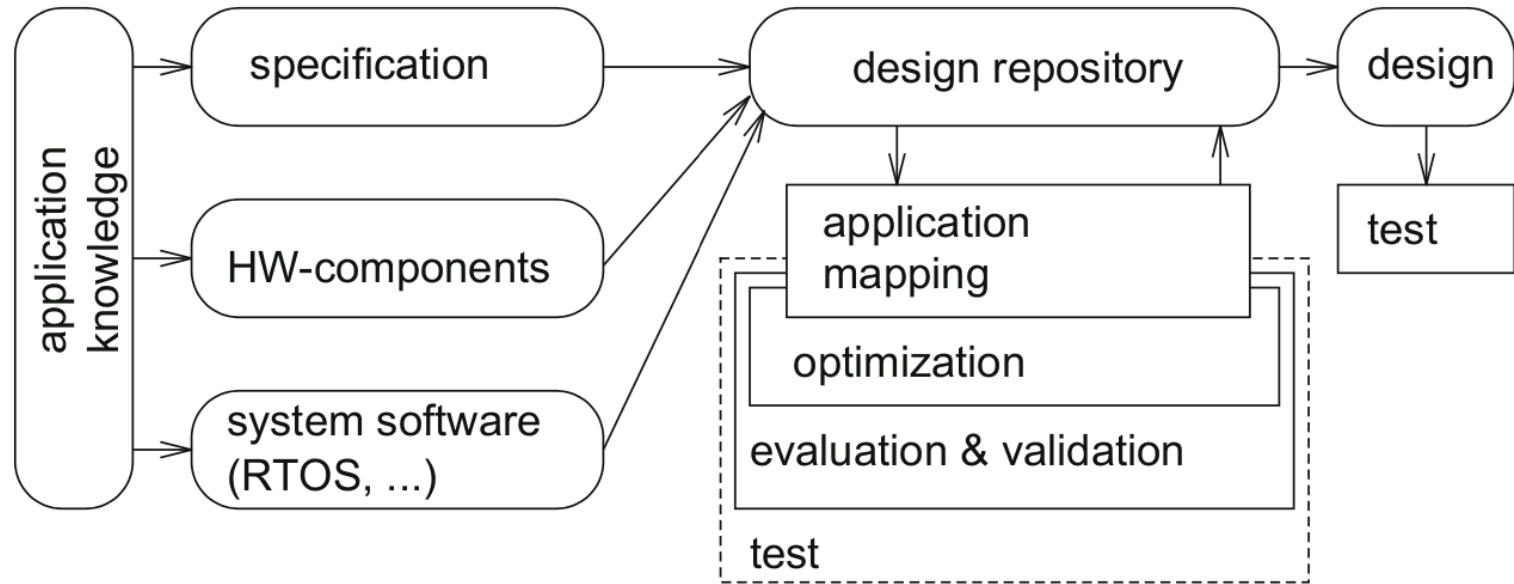


Fig. 1.8 Simplified design information flow

Marwedel, Peter. *Embedded System Design*. Springer, 2018.

Design Flow Unfolded

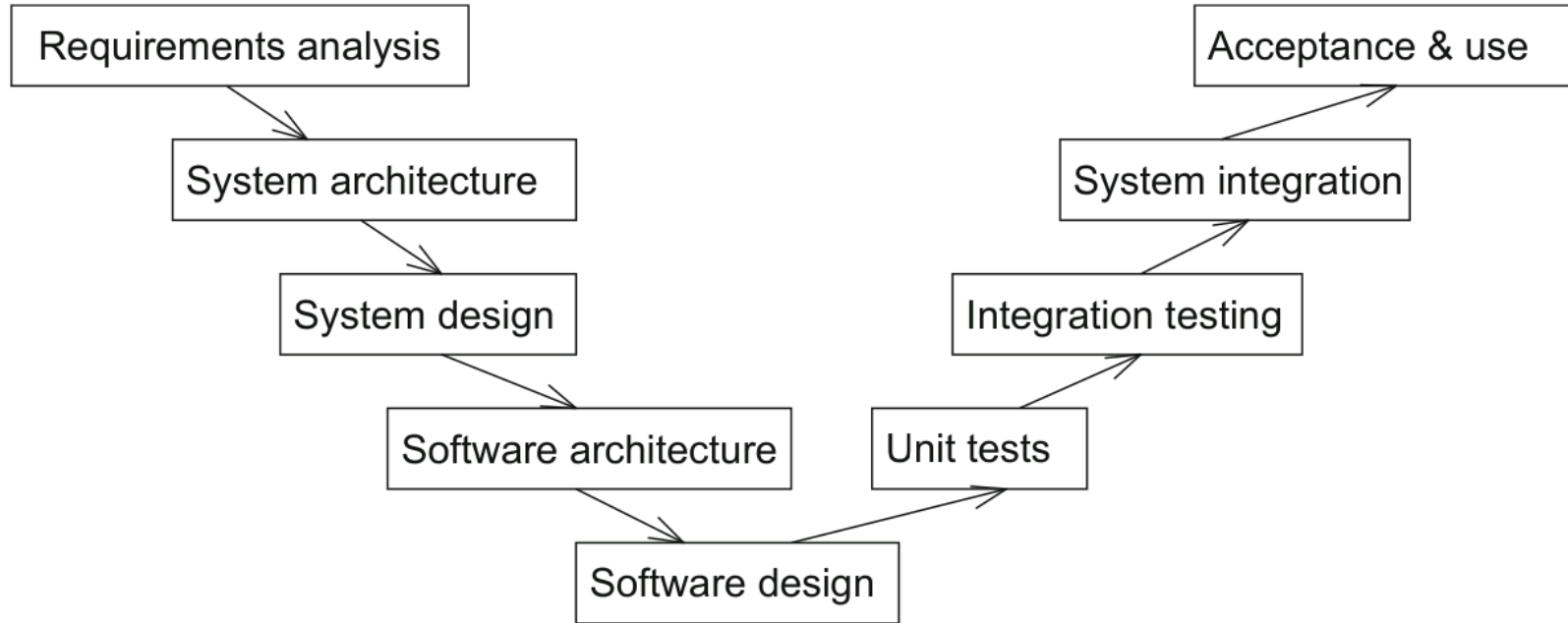
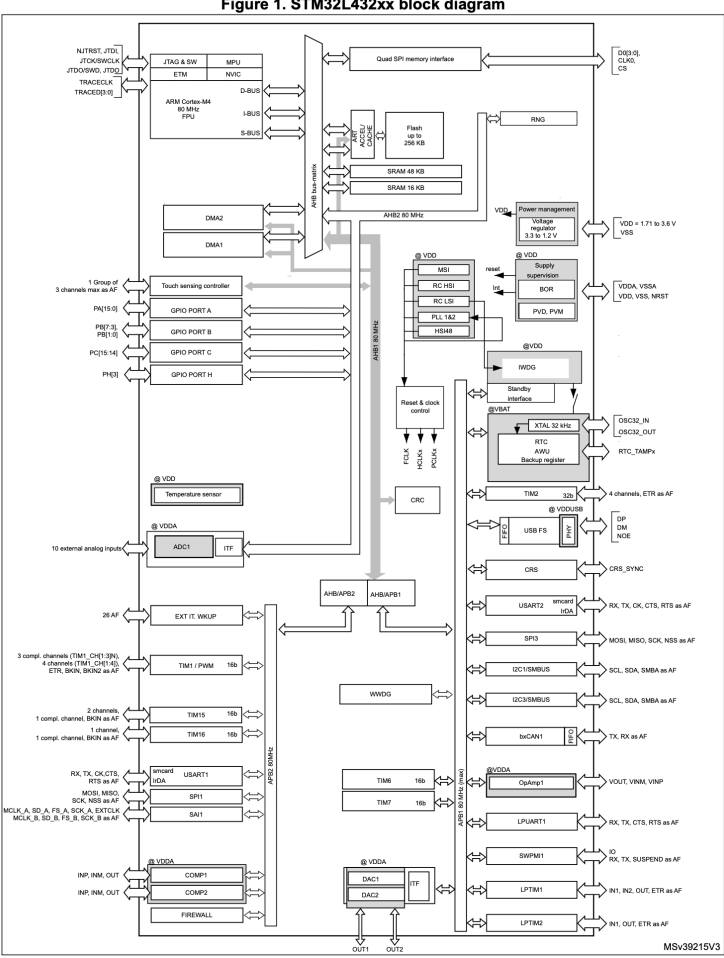
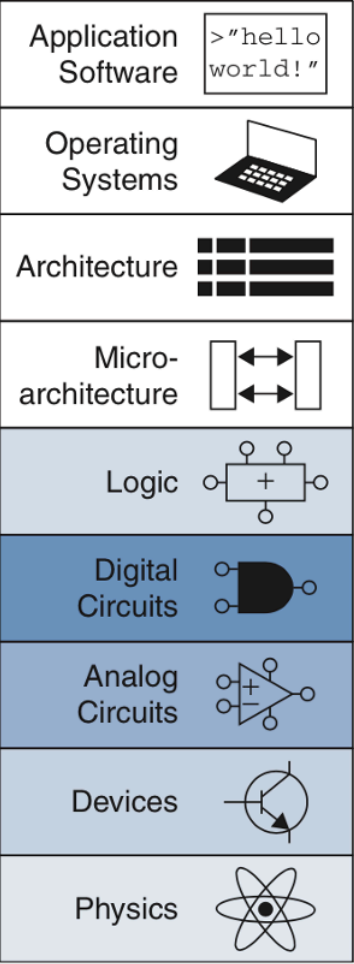


Fig. 1.10 Design flow for the V-model

Marwedel, Peter. *Embedded System Design*. Springer, 2018.

MicroPs in Context



From DDCA ARM edition

STM32L432KC Datasheet - p. 13

What is an FPGA?

- FPGA stands for **Field Programmable Gate Array**.
- Programmed by configuring **Logic Cells (LC)** and connecting them.

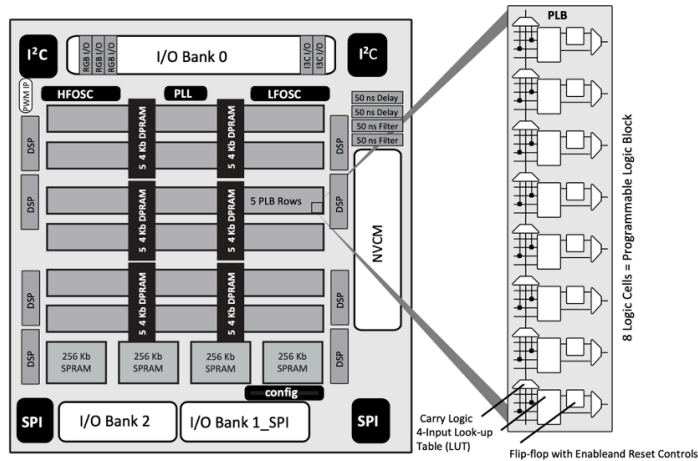


Figure 3.1. ICE40UP5K Device, Top View

ice40 UltraPlus Family Data Sheet p. 11

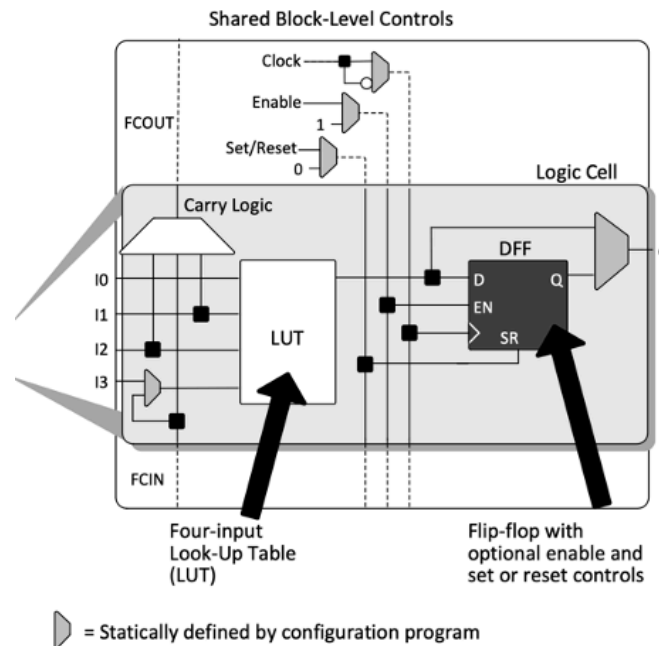


Figure 3.2. PLB Block Diagram

ice40 UltraPlus Family Data Sheet p. 12



UPduino
v3.1

What is an MCU?

- MCU stands for **Microcontroller Unit**
- Comprised of a processor (think E85) surrounded by a bunch of peripherals.
 - Serial communication: UART, SPI
 - Timers & Pulse Width Modulation (PWM)
 - General-purpose Input-Output (GPIO) Controllers

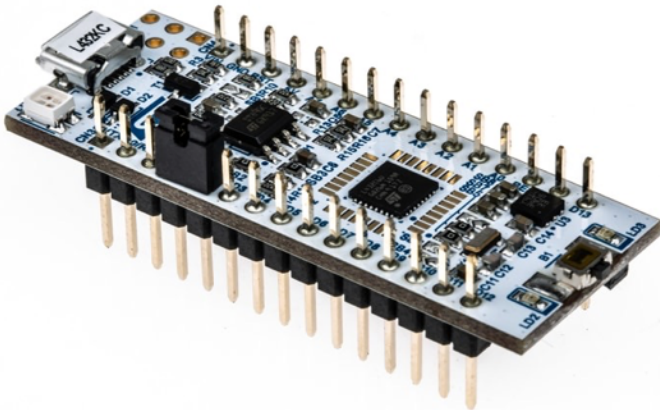
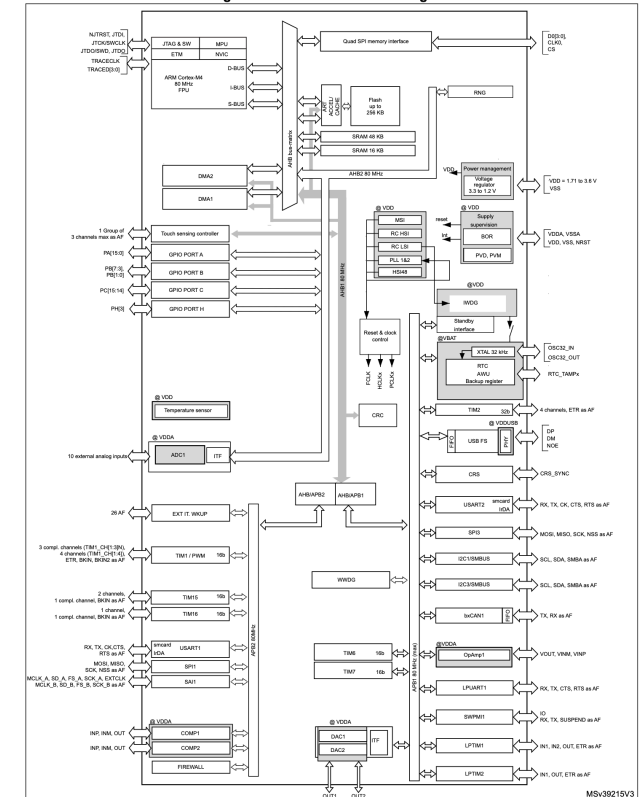


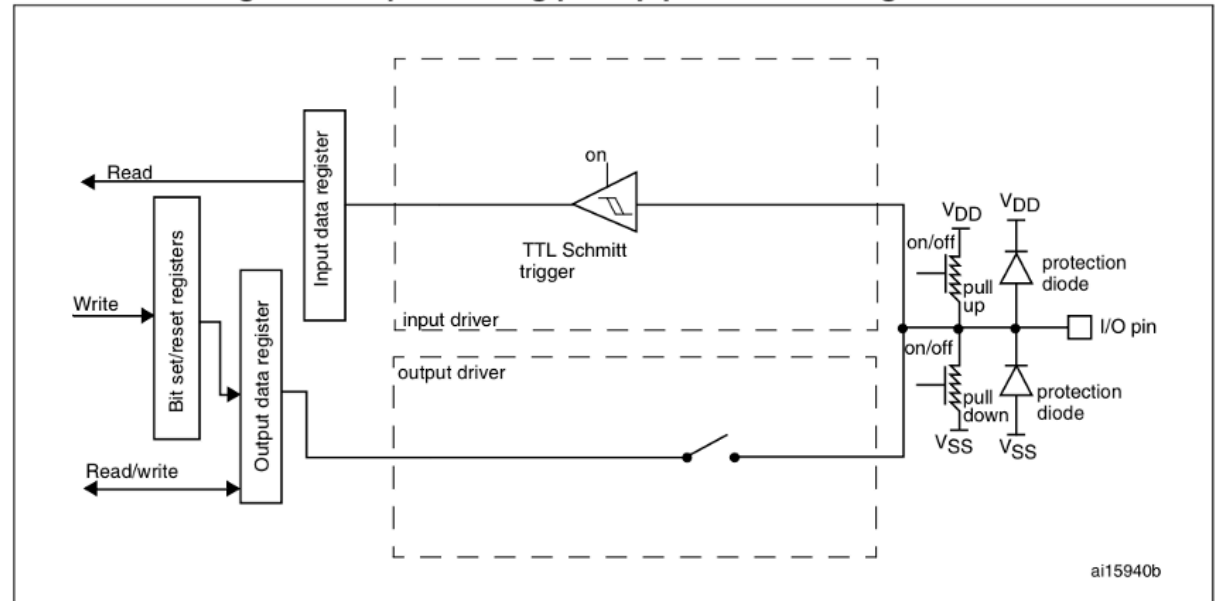
Figure 1. STM32L432xx block diagram



MicroPs in Context

- Focus on learning general concepts – specifics are obsolete tomorrow!
- Some core concepts endure
 - Understanding device layout and how to read documentation
 - Basic configuration/programming flow
 - Memory-mapped I/O
 - Peripheral configuration
 - Clock configuration
 - Peripherals
 - UART
 - SPI
 - Timers
 - PWM

Figure 18. Input floating/pull up/pull down configurations

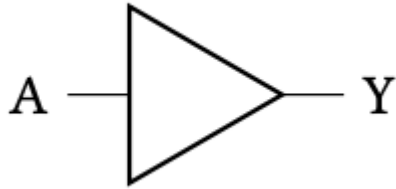


Technical Kickoff: E85 Warp Speed Review

Boolean Logic & Verilog Idiom Review

- BUF
- NOT
- AND
- OR
- NAND
- NOR
- XOR
- XNOR
- AND3
- 2:1 Mux
- 2:4 Decoder

BUF



Truth Table

A	Y
0	0
1	1

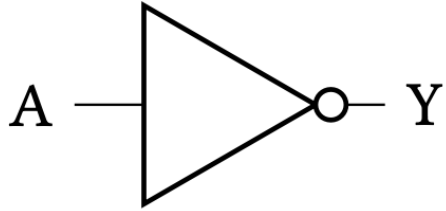
Boolean Expression

$$Y = A$$

Verilog Idiom

```
assign y = a;
```

NOT



Truth Table

A	Y
0	1
1	0

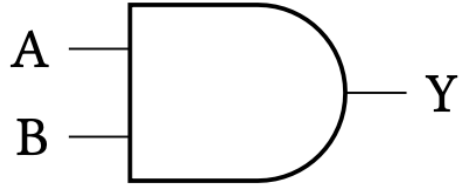
Boolean Expression

$$Y = \bar{A}$$

Verilog Idiom

```
assign y = ~a;
```

AND



Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

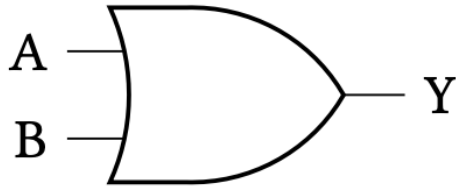
Boolean Expression

$$Y = AB$$

Verilog Idiom

```
assign y = a & b;
```


OR



Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

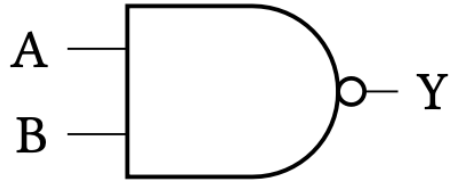
Boolean Expression

$$Y = A + B$$

Verilog Idiom

```
assign y = a | b;
```

NAND



Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

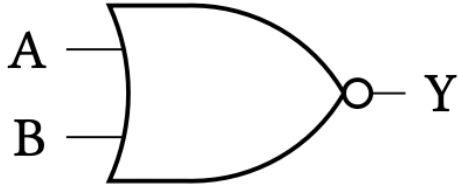
Boolean Expression

$$Y = \overline{AB}$$

Verilog Idiom

```
assign y = ~(a & b);
```

NOR



Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

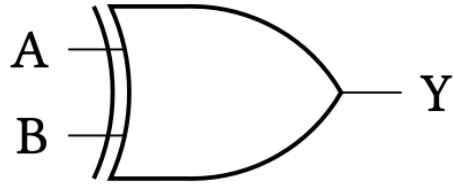
Boolean Expression

$$Y = \overline{A + B}$$

Verilog Idiom

```
assign y = ~(a | b);
```

XOR



Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

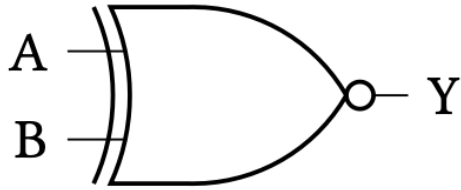
Boolean Expression

$$Y = A \oplus B$$

Verilog Idiom

```
assign y = a ^ b;
```

XNOR



Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

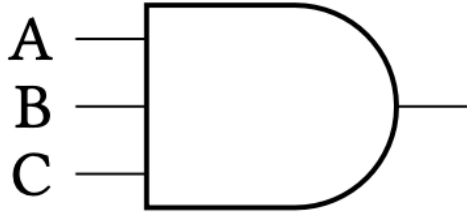
Boolean Expression

$$Y = \overline{A \oplus B}$$

Verilog Idiom

```
assign y = ~(a ^ b);
```

AND3



Truth Table

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

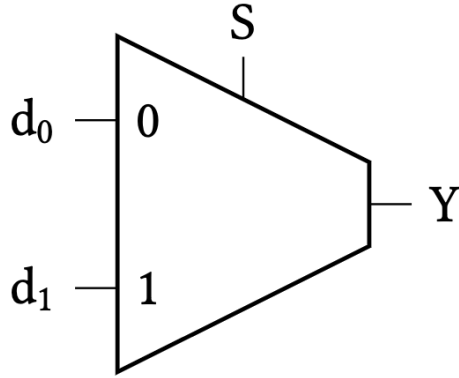
Boolean Expression

$$Y = ABC$$

Verilog Idiom

```
assign y = a & b & c;
```

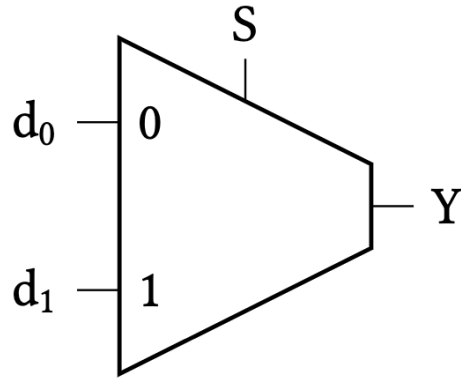
2:1 Multiplexer (MUX)



Verilog Idiom

```
assign y = s ? d1 : d0;
```

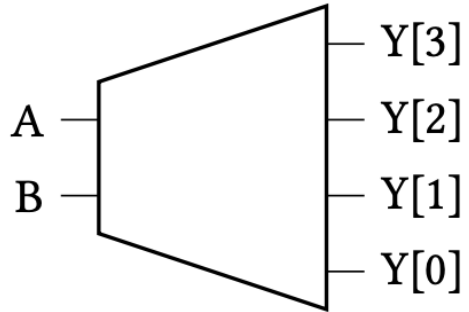
2:1 Multiplexer (MUX)



Truth Table

S	d_1	d_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0

2:4 Decoder



Truth Table

A	B	Y3	Y2	Y1	Y0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Verilog Idiom

```
1 always_comb
2     case(s)
3         2'b00: y = 4'b0001;
4         2'b01: y = 4'b0010;
5         2'b10: y = 4'b0100;
6         2'b11: y = 4'b1000;
7         default: y = 4'bxxxx;
```

Analog Behavior of Digital Systems

DC Logic Gate Transfer Characteristics

- VIH - Lowest **input** voltage recognized as logical 1
- VIL - Highest **input** voltage recognized as logical 0
- VOH - Lowest **output** voltage indicating logical 1
- VOL - Highest **output** voltage indicating logical 0
- Noise margins - **Difference between the high and low values for the input or output levels.**

DC Logic Gate Transfer Characteristics

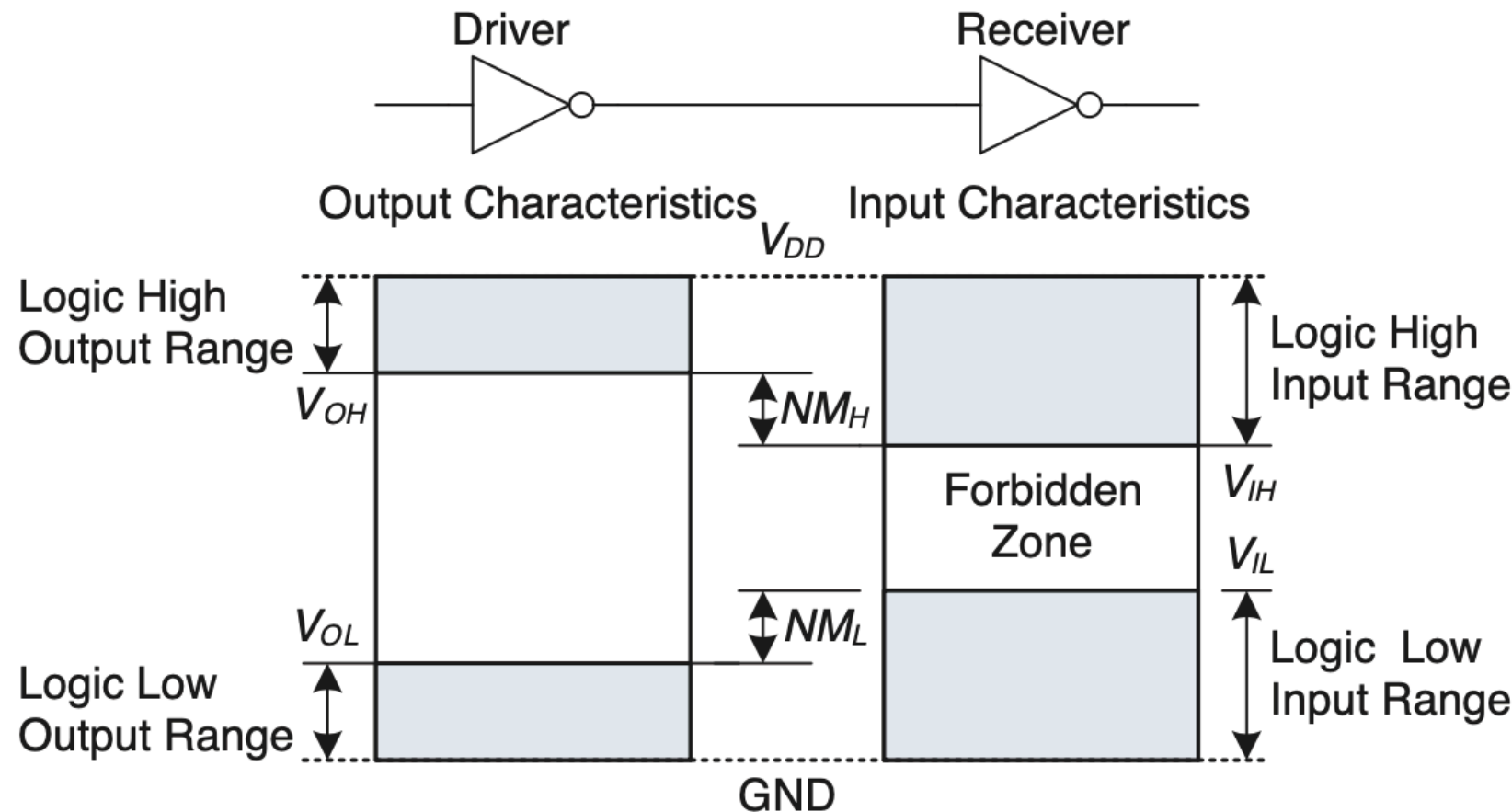


Figure 1.23 from *Digital Design and Computer Architecture: RISC-V Edition* by Harris & Harris

Important Specs

- I_{in} – Input leakage current (a current flowing in or out of the pin toward a power rail. This can cause invalid logic levels if you're not careful and use too big of a pulldown/up resistor.)
- I_{out} – Maximum output driving current
- C_{in} – Input pin capacitance
- I_{DD} – Supply current. DD stands for from drain to drain.

Wrap Up

- Get kit and start on Lab 1 ASAP
- Also, checkout breadboard from Sam to use for the semester.
- Schedule checkoff time (try to stay within your scheduled section. Help each other out if others need to swap).
- Lab demos this week during the first hour of the lab slot
 - Soldering refresh
 - Basic FPGA and MCU programming (see tutorials on the website)